

Table 1: Results of test examples 1-3

N	Pdim	DFPRP				MPRP1				RESHYBRIDSCG				RDFPM1				RDFPM2				DFDPP				
		IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM	
1	5000	x_1^1	1	3	0.0077	0	67	135	0.0847	7.40E-11	21	43	0.0329	7.75E-11	11	17	0.0210	0	4	10	0.0104	7.95E-21	1	3	0.0389	0
	5000	x_2^1	1	3	0.0038	0	72	146	0.1289	9.24E-11	21	45	0.0285	4.97E-11	25	33	0.0339	6.49E-11	10	18	0.0170	1.30E-20	1	3	0.0112	0
	5000	x_3^1	1	3	0.0039	0	85	184	0.0967	8.89E-11	17	37	0.0233	5.42E-11	13	18	0.0190	6.88E-24	2	4	0.0052	0	4	8	0.0106	0
	5000	x_4^1	1	3	0.0037	0	74	149	0.0844	8.79E-11	21	43	0.0285	7.85E-11	17	23	0.0249	0	12	19	0.0187	2.94E-23	1	3	0.0033	0
	5000	x_5^1	1	3	0.0037	0	56	113	0.0653	8.48E-11	21	43	0.0272	7.70E-11	14	19	0.0209	1.11E-20	8	16	0.0160	2.57E-22	1	3	0.0036	0
	5000	x_6^1	1	3	0.0037	0	74	149	0.0870	8.80E-11	21	43	0.0305	7.85E-11	23	28	0.0298	0	12	19	0.0191	2.93E-23	1	3	0.0042	0
	10000	x_1^1	1	3	0.0052	0	67	135	0.1140	7.40E-11	21	45	0.0430	4.38E-11	6	12	0.0175	0	5	12	0.0168	1.68E-21	1	3	0.0045	0
	10000	x_2^1	1	3	0.0045	0	73	148	0.1321	9.32E-11	21	45	0.0443	7.03E-11	29	36	0.0578	8.72E-11	12	21	0.0305	4.29E-22	1	3	0.0046	0
	10000	x_3^1	1	3	0.0049	0	86	186	0.1507	8.96E-11	17	37	0.0357	5.42E-11	13	18	0.0367	6.88E-24	2	4	0.0061	0	4	8	0.0119	0
	10000	x_4^1	1	3	0.0043	0	75	151	0.1277	8.87E-11	21	45	0.0425	4.44E-11	33	37	0.0625	9.73E-11	13	21	0.0317	0	1	3	0.0044	0
	10000	x_5^1	1	3	0.0045	0	57	115	0.0906	8.55E-11	21	45	0.0469	4.36E-11	16	24	0.0385	2.63E-11	7	14	0.0197	0	1	3	0.0047	0
	10000	x_6^1	1	3	0.0049	0	75	151	0.1270	8.88E-11	21	45	0.0440	4.44E-11	20	27	0.0426	0	12	19	0.0289	4.11E-23	1	3	0.0044	0
	50000	x_1^1	1	3	0.0133	0	67	135	0.4491	7.40E-11	21	45	0.1609	9.80E-11	7	12	0.0747	3.70E-21	3	8	0.0386	3.57E-18	1	3	0.0127	0
	50000	x_2^1	1	3	0.0147	0	76	154	0.5108	7.57E-11	22	47	0.1646	4.40E-11	27	34	0.2024	9.25E-11	11	18	0.1001	8.28E-18	1	3	0.0143	0
	50000	x_3^1	1	3	0.0131	0	89	192	0.6107	7.28E-11	17	37	0.1281	5.42E-11	13	18	0.1062	6.88E-24	2	4	0.0252	0	4	8	0.0446	0
	50000	x_4^1	1	3	0.0125	0	78	157	0.5305	7.21E-11	21	45	0.1549	9.83E-11	20	27	0.1605	1.10E-23	11	18	0.1020	1.05E-20	1	3	0.0130	0
	50000	x_5^1	1	3	0.0136	0	59	119	0.3996	9.74E-11	21	45	0.1560	9.74E-11	9	15	0.0851	1.42E-18	7	13	0.0711	2.84E-18	1	3	0.0137	0
	50000	x_6^1	1	3	0.0128	0	78	157	0.5470	7.21E-11	21	45	0.1520	9.86E-11	18	24	0.1439	1.06E-22	11	18	0.0997	2.07E-21	1	3	0.0133	0
2	5000	x_1^1	1	3	0.0046	0	63	131	0.0650	9.27E-11	19	41	0.0223	9.86E-11	13	21	0.0277	0	5	13	0.0091	0	5	9	0.0086	5.95E-11
	5000	x_2^1	1	4	0.0039	0	82	171	0.0842	9.26E-11	2	6	0.0047	0	27	35	0.0357	0	10	16	0.0141	0	1	4	0.0038	0
	5000	x_3^1	1	3	0.0036	2.22E-16	76	167	0.0841	9.12E-11	17	35	0.0213	7.89E-11	30	33	0.0331	5.60E-11	7	15	0.0109	8.18E-11	6	12	0.0102	5.11E-13
	5000	x_4^1	3	6	0.0055	0	82	170	0.0878	7.48E-11	23	50	0.0263	4.62E-11	38	44	0.0491	7.96E-11	17	27	0.0223	7.45E-12	5	9	0.0079	0
	5000	x_5^1	1	3	0.0035	0	78	159	0.0883	8.72E-11	20	43	0.0235	9.34E-11	37	42	0.0465	6.23E-11	8	12	0.0112	0	1	3	0.0033	0
	5000	x_6^1	3	6	0.0057	0	81	168	0.0840	9.95E-11	23	50	0.0262	5.00E-11	38	44	0.0458	8.14E-11	17	27	0.0219	7.35E-12	5	9	0.0090	0
	10000	x_1^1	1	3	0.0038	0	63	131	0.0978	9.34E-11	20	41	0.0344	9.77E-11	5	13	0.0144	0	5	13	0.0136	0	5	9	0.0118	5.96E-11
	10000	x_2^1	1	4	0.0051	0	83	173	0.1380	9.35E-11	2	6	0.0056	0	29	36	0.0520	0	10	16	0.0231	0	1	4	0.0057	0
	10000	x_3^1	1	3	0.0044	2.22E-16	77	169	0.1246	9.20E-11	17	35	0.0292	7.89E-11	30	33	0.0467	5.60E-11	7	15	0.0151	8.18E-11	6	12	0.0143	7.22E-13
	10000	x_4^1	3	6	0.0081	0	83	172	0.1284	7.45E-11	23	50	0.0381	6.63E-11	36	42	0.0637	5.33E-11	18	28	0.0327	5.63E-12	5	9	0.0109	0
	10000	x_5^1	1	3	0.0047	0	79	161	0.1275	8.80E-11	21	43	0.0345	9.25E-11	23	29	0.0420	0	27	42	0.0512	0	1	3	0.0045	0
	10000	x_6^1	3	6	0.0089	0	83	172	0.1360	7.25E-11	23	50	0.0396	6.90E-11	39	44	0.0638	9.73E-11	18	28	0.0347	5.67E-12	5	9	0.0110	0
	50000	x_1^1	1	3	0.0109	0	63	131	0.3748	9.40E-11	20	43	0.1409	8.74E-11	11	18	0.0832	0	5	13	0.0478	0	5	9	0.0395	5.96E-11
	50000	x_2^1	1	4	0.0134	0	86	179	0.5109	7.60E-11	2	6	0.0168	0	26	34	0.1714	0	13	23	0.1001	0	1	4	0.0141	0
	50000	x_3^1	1	3	0.0114	2.22E-16	80	175	0.5556	7.48E-11	17	35	0.1198	7.89E-11	30	33	0.1650	5.60E-11	7	15	0.0573	8.18E-11	6	12	0.0543	1.61E-12
	50000	x_4^1	3	6	0.0253	0	85	176	0.8151	8.39E-11	24	52	0.1478	3.43E-11	41	46	0.2328	5.89E-11	18	28	0.1272	2.04E-11	5	9	0.0410	0
	50000	x_5^1	1	3	0.0124	0	82	167	0.4907	7.15E-11	21	45	0.1333	8.27E-11	31	37	0.1950	9.83E-11	28	42	0.2025	0	1	3	0.0118	0
	50000	x_6^1	3	6	0.0264	0	85	176	0.5812	8.35E-11	24	52	0.1574	3.46E-11	41	46	0.2343	5.89E-11	18	28	0.1285	2.05E-11	5	9	0.0377	0
3	5000	x_1^1	1	4	0.0050	0	75	376	0.1340	9.32E-11	12	40	0.0203	4.61E-11	1	6	0.0150	0	1	6	0.0049	0	5	11	0.0100	0
	5000	x_2^1	3	8	0.0072	0	89	448	0.1610	8.89E-11	14	44	0.0210	5.99E-11	74	140	0.1142	9.41E-11	14	49	0.0326	1.17E-11	1	4	0.0046	0
	5000	x_3^1	5	12	0.0093	4.44E-16	41	216	0.0797	7.65E-11	14	42	0.0217	1.78E-11	41	46	0.0523	4.13E-11	10	45	0.0260	1.62E-11	8	16	0.0147	0
	5000	x_4^1	1	4	0.0038	0	86	432	0.1558	8.03E-11	16	50	0.0238	5.32E-11	69	125	0.1098	6.40E-11	21	59	0.0413	2.03E-11	1	4	0.0038	0
	5000	x_5^1	1	4	0.0045	0	78	392	0.1439	8.80E-11	14	43	0.0218	5.10E-11	58	105	0.0918	6.46E-11	9	41	0.0247	8.67E-11	1	4	0.0041	0
	5000	x_6^1	1	4	0.0040	0	86	432	0.1568	8.06E-11	16	50	0.0236	5.33E-11	53	102	0.0883	7.35E-11	17	56	0.0392	1.28E-11	1	4	0.0038	0
	10000	x_1^1	1	4	0.0050	0	75	376	0.2040	9.32E-11	12	40	0.0300	6.35E-11	1	6	0.0063	0	1	6	0.0064	0	5	11	0.0145	0
	10000	x_2^1	3	8	0.0113	4.44E-16	90	453	0.2483	8.91E-11	14	44	0.0339	8.45E-11	55	114	0.1485	9.88E-11	14	49	0.0545	8.20E-12	1	4	0.0054	0
	10000	x_3^1	5	12	0.0145	4.44E-16	5	53	0.0298	0	14	42	0.0341	1.78E-11	41	46	0.0785	4.13E-11	10	45	0.0401	1.62E-11	8	16	0.0221	0
	10000	x_4^1	1	4	0.0050	0	87	437	0.2446	8.43E-11	16	50	0.0393	7.53E-11	77	149	0.1850	9.78E-								

Table 3: Results of test examples 7-8

N	Pdim	DFPRP			MPRPI			RESHYBRIDSCG			RDPPI			RDFPM2			DFDFP									
		IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM	IN	FVAL	PTIME	NORM					
7	5000	x_1^4	8	15	0.0196	2.22E-12	65	308	0.1337	9.30E-11	14	46	0.0306	4.12E-11	34	0.0397	6.75E-11	11	49	0.0360	5.38E-11	13	27	0.0265	1.11E-11	
	5000	x_2^2	10	18	0.0188	6.04E-11	62	307	0.1423	9.11E-11	17	51	0.0358	7.03E-11	25	56	0.0570	7.32E-11	15	60	0.0515	3.01E-11	12	29	0.0253	2.06E-11
	5000	x_3^4	13	24	0.0247	6.16E-12	70	341	0.1394	8.67E-11	16	52	0.0341	9.93E-11	37	70	0.0835	8.46E-11	14	58	0.0436	5.12E-11	13	27	0.0266	5.01E-11
	5000	x_4^4	17	30	0.0346	1.91E-12	66	330	0.1398	9.85E-11	16	52	0.0343	4.32E-11	31	74	0.0818	4.65E-11	16	63	0.0508	3.87E-11	13	27	0.0262	5.23E-11
	5000	x_5^4	13	22	0.0238	8.99E-11	65	325	0.1328	9.90E-11	16	52	0.0340	3.54E-11	34	67	0.0818	7.37E-11	17	63	0.0506	3.89E-11	12	25	0.0243	6.36E-11
	5000	x_6^4	17	30	0.0312	1.75E-12	66	330	0.1394	9.86E-11	16	52	0.0342	4.31E-11	31	78	0.0808	6.42E-11	16	63	0.0506	3.89E-11	13	27	0.0262	5.22E-11
	10000	x_1^4	8	15	0.0255	3.14E-12	62	300	0.2088	9.92E-11	14	46	0.0488	5.83E-11	11	34	0.0498	9.51E-11	11	49	0.0583	7.60E-11	13	27	0.0412	1.58E-11
	10000	x_2^2	10	18	0.0290	8.54E-11	63	312	0.1980	8.57E-11	17	51	0.0573	9.95E-11	19	52	0.0805	3.56E-11	15	57	0.0727	8.01E-11	12	29	0.0398	3.20E-11
	10000	x_3^4	17	30	0.0510	7.30E-12	71	346	0.2229	8.15E-11	17	52	0.0559	8.24E-11	30	64	0.1113	7.46E-11	15	61	0.0773	4.86E-11	13	27	0.0398	7.09E-11
	10000	x_4^4	17	30	0.0530	2.64E-12	67	335	0.2398	9.26E-11	16	52	0.0618	6.10E-11	34	75	0.1712	6.39E-11	18	65	0.0895	6.45E-11	13	27	0.0416	7.39E-11
	10000	x_5^4	13	24	0.0397	9.78E-13	66	330	0.2074	9.31E-11	16	52	0.0548	5.01E-11	31	68	0.1343	9.50E-11	19	65	0.0850	3.32E-11	12	25	0.0439	9.00E-11
	10000	x_6^4	17	30	0.0527	2.53E-12	67	335	0.2265	9.27E-11	16	52	0.0542	6.10E-11	38	84	0.1606	4.77E-11	18	65	0.0889	6.06E-11	13	27	0.0395	7.39E-11
	50000	x_1^4	8	15	0.1056	7.01E-12	62	306	0.7964	7.34E-11	15	46	0.2002	7.64E-11	12	31	0.1944	5.80E-11	12	53	0.2862	2.55E-11	13	27	0.1672	3.53E-11
	50000	x_2^2	10	20	0.1258	1.47E-12	65	322	0.8268	8.47E-11	17	54	0.2402	7.25E-11	21	53	0.3287	8.71E-11	16	61	0.3144	2.69E-11	12	29	0.1789	7.14E-11
	50000	x_3^4	18	30	0.1884	5.02E-11	73	356	0.9277	8.06E-11	17	55	0.2349	6.01E-11	28	68	0.4228	9.44E-11	15	58	0.3008	2.19E-11	14	29	0.1816	1.40E-11
50000	x_4^4	17	30	0.2065	5.81E-12	69	345	0.9312	9.16E-11	17	52	0.2196	8.00E-11	30	70	0.4638	5.87E-11	19	69	0.3723	2.19E-11	14	29	0.1846	1.47E-11	
50000	x_5^4	13	24	0.1522	2.19E-12	68	340	0.8824	9.21E-11	17	52	0.2421	6.57E-11	28	63	0.4261	9.65E-11	22	69	0.3825	1.79E-11	13	27	0.1739	1.78E-11	
50000	x_6^4	17	30	0.2020	5.76E-12	69	345	0.9193	9.16E-11	17	52	0.2305	8.00E-11	44	90	0.6603	7.39E-11	19	69	0.3892	2.20E-11	14	29	0.1961	1.47E-11	
8	5000	x_1^4	11	23	0.0356	8.65E-11	31	81	0.0591	4.89E-11	20	44	0.0389	4.99E-11	***	***	***	***	***	***	***	***	14	26	0.0355	2.41E-11
	5000	x_2^2	11	25	0.0293	8.60E-11	30	77	0.0576	1.00E-10	19	41	0.0348	7.17E-11	***	***	***	***	***	***	***	***	14	27	0.0316	1.49E-11
	5000	x_3^4	12	25	0.0320	4.63E-11	32	86	0.0614	9.90E-11	19	42	0.0435	2.98E-11	***	***	***	***	***	***	***	***	13	25	0.0350	9.74E-11
	5000	x_4^4	11	25	0.0297	7.91E-11	31	83	0.0575	8.10E-11	20	42	0.0368	9.35E-11	***	***	***	***	***	***	***	***	14	27	0.0315	2.89E-11
	5000	x_5^4	11	23	0.0281	7.22E-11	35	88	0.0648	7.52E-11	18	40	0.0348	9.46E-11	***	***	***	***	***	***	***	***	13	25	0.0306	5.90E-11
	5000	x_6^4	11	25	0.0291	4.86E-11	37	97	0.0719	9.24E-11	19	42	0.0386	4.11E-11	***	***	***	***	***	***	***	***	13	25	0.0299	6.64E-11
	10000	x_1^4	12	27	0.0500	6.04E-11	36	92	0.1210	1.00E-11	21	46	0.0666	4.70E-11	***	***	***	***	***	***	***	***	15	28	0.0595	1.75E-11
	10000	x_2^2	11	23	0.0441	6.57E-11	31	77	0.0922	5.49E-11	18	40	0.0570	8.89E-11	***	***	***	***	***	***	***	***	14	27	0.0535	1.54E-11
	10000	x_3^4	12	27	0.0537	3.16E-11	37	87	0.1095	4.92E-11	22	47	0.0751	4.70E-11	***	***	***	***	***	***	***	***	11	22	0.0425	8.63E-11
	10000	x_4^4	12	27	0.0524	8.13E-11	38	93	0.1212	8.92E-12	20	44	0.0634	3.06E-11	***	***	***	***	***	***	***	***	13	26	0.0502	1.82E-11
	10000	x_5^4	12	27	0.0534	6.58E-11	34	95	0.1076	8.57E-11	18	40	0.0577	5.61E-11	***	***	***	***	***	***	***	***	17	30	0.0746	3.17E-11
	10000	x_6^4	12	27	0.0473	7.45E-11	46	114	0.1470	1.44E-11	21	46	0.0708	2.57E-11	***	***	***	***	***	***	***	***	14	27	0.0544	1.06E-12
	50000	x_1^4	14	29	0.2371	3.13E-11	44	104	0.5748	6.22E-11	20	44	0.2871	9.14E-11	***	***	***	***	***	***	***	***	16	30	0.2932	9.82E-12
	50000	x_2^2	13	29	0.2279	9.21E-11	57	129	0.7292	2.98E-11	21	45	0.2965	9.16E-11	***	***	***	***	***	***	***	***	15	28	0.2866	2.49E-11
	50000	x_3^4	14	29	0.2464	3.77E-11	35	92	0.4736	8.80E-11	20	42	0.2591	7.21E-11	***	***	***	***	***	***	***	***	16	30	0.2874	2.33E-11
	50000	x_4^4	14	29	0.2315	4.56E-11	42	110	0.5665	8.14E-11	21	46	0.2911	5.49E-11	***	***	***	***	***	***	***	***	13	25	0.2288	6.79E-11
	50000	x_5^4	14	29	0.2349	3.69E-11	55	145	0.7529	3.26E-11	20	42	0.2650	8.58E-11	***	***	***	***	***	***	***	***	16	29	0.2684	3.23E-11
	50000	x_6^4	14	29	0.2395	4.24E-11	34	89	0.4885	2.60E-11	21	46	0.2883	5.72E-11	***	***	***	***	***	***	***	***	16	29	0.2670	2.76E-11